

Introduction

Functions are critical to the study of mathematical objects, and groups are no different. The study of functions can reveal how groups interact with one another, but they can also tell us when *two groups are the same!* For example, you may have realized in the previous reading that many of the groups we were working with looked similar. Below are the tables describing the binary operations for $\mathbb{Z}_2$ under addition, $U = \{[1]_6, [5]_6\}$ under multiplication mod 6, and the cyclic subgroup $\langle [2]_4 \rangle$ of $\mathbb{Z}_4$.

$+$	$[0]_2$	$[1]_2$
$[0]_2$	$[0]_2$	$[1]_2$
$[1]_2$	$[1]_2$	$[0]_2$
$\mathbb{Z}_2$		

$\times$	$[1]_6$	$[5]_6$
$[1]_6$	$[1]_6$	$[5]_6$
$[5]_6$	$[5]_6$	$[1]_6$
U		

$+$	$[0]_4$	$[2]_4$
$[0]_4$	$[0]_4$	$[2]_4$
$[2]_4$	$[2]_4$	$[0]_4$
$\langle 2 \rangle \subseteq \mathbb{Z}_4$		

These are all the same table, which means these are all the same group (in disguise)! The only thing that changed was the name of the elements.

Learning Objectives

- Define a group homomorphism, and determine whether a function is a homomorphism or not.
- Define the kernel and image of a homomorphism. Determine and prove the kernel and image of a given homomorphism.
- Prove simple properties about group homomorphisms.
- Define a group isomorphism, and what it means to be isomorphic. Prove that a function is an isomorphism, and prove whether basic groups are isomorphic or not.
- Prove simple properties about isomorphisms and isomorphic groups.

Homomorphisms

While we've already learned about functions between sets $f : A \rightarrow B$, our sets now have *additional structure* in the form of their binary operators. Thus we must also impose an additional structure on our functions; namely, that they map the group operation from the domain into the codomain.

Definition 1

Given two groups $(G, \cdot)$ and $(H, \star)$, a function $\phi : G \rightarrow H$ is said to be a *group homomorphism* if for all $x, y \in G$, $\phi(x \cdot y) = \phi(x) \star \phi(y)$.

When we're lazy and the group operations are understood, we again drop the group operations and just write $\phi(xy) = \phi(x)\phi(y)$. But remember, there are two potentially different group operations at play in this equation.

Example 2

Determine whether the map $\phi : \mathbb{Z}_6 \rightarrow \mathbb{Z}_{12}$ given by $\phi([x]_6) = [2x]_{12}$ is a group homomorphism.

Solution. We must first verify that ϕ is well defined; namely, that if $[a]_6 = [b]_6$ then $\phi([a]_6) = \phi([b]_6)$. Suppose that $[a]_6 = [b]_6$ so that $a - b = 6k$ for some $k \in \mathbb{Z}$. Now $\phi([a]_6) - \phi([b]_6) = 2[a]_6 - 2[b]_6 = 2([a]_6 - [b]_6) = 12k$, showing that $\phi([a]_6) \equiv \phi([b]_6) \pmod{12}$.

Finally, to show that ϕ preserves the group structure, note that if $[a]_6, [b]_6 \in \mathbb{Z}_6$ then

$$\phi([a]_6 + [b]_6) = \phi([a + b]_6) = [2(a + b)]_{12} = [2a]_{12} + [2b]_{12} = \phi([a]_6) + \phi([b]_6),$$

which is exactly what we wanted to show. ■

If the “well-defined” argument above is bothering you, remember that ϕ is a map on *equivalence classes*. Thus we need to show that, regardless of which member of the equivalence class we use to compute ϕ , we would get the same answer out. For example, pretend that we wanted to define $\zeta : \mathbb{Z}_6 \rightarrow \mathbb{Z}_{12}$ by means of $\zeta([x]_6) = [3x]_{12}$. This map is **not well-defined**. Indeed, note that $[1]_6 = [7]_6$, but that $\phi([1]_6) = [3]_{12}$ while $\phi([7]_6) = [9]_{12}$, and these are not equal. By the same token, the map $\xi : \mathbb{Z}_6 \rightarrow \mathbb{Z}_{12}, \xi([x]_6) = [x]_{12}$ is well-defined, but is not a group homomorphism. You can check the well-definedness of the map yourself, but to see that it's not a homomorphism, note that $\xi([3]_6 + [4]_6) = \xi([1]_6) = [1]_{12}$, but $\xi([4]_6) + \xi([3]_6) = [7]_{12}$.

Definition 3

Given a group homomorphism $\phi : G \rightarrow H$, we define the

- *kernel of ϕ* as $\ker(\phi) = \{g \in G : \phi(g) = e_H\}$, where e_H is the identity element in H ;
- *image of ϕ* as $\text{im}(\phi) = \phi(G) = \{\phi(g) : g \in G\}$, which is our usual notion of the image of a function.

Example 4

Consider the map $\phi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_6$ given by $\phi([x]_{12}) = [2x]_6$. Determine the kernel and image of this map.

Solution. Keep in mind that this map is not the same as the one given in Example 2, as the domain and codomain have switched. We leave it as an exercise to check that this is a well-defined homomorphism. Since the domain here is rather small, we can just compute the output of every possible element, as shown in the table below:

x	0	1	2	3	4	5	6	7	8	9	10	11
$2x \pmod{6}$	0	2	4	0	2	4	0	2	4	0	2	4

Evidently, $\ker(\phi) = \{[0]_{12}, [3]_{12}, [6]_{12}, [9]_{12}\}$ and $\text{im}(\phi) = \{[0]_6, [2]_6, [4]_6\}$. ■

The additional structure afforded by a group homomorphism allows us to make a lot of powerful statements that would otherwise not be true for generic functions.

Proposition 5

Suppose that $\phi : G \rightarrow H$ is a group homomorphism.

1. $\phi(e_G) = e_H$, where e_G and e_H are the identity elements of G and H respectively.
2. For any $n \in \mathbb{Z}$, $\phi(g^n) = \phi(g)^n$. Notably, $\phi(a^{-1}) = \phi(a)^{-1}$.
3. For any $g \in G$ such that $|g| < \infty$, $|\phi(g)|$ divides $|g|$.
4. $\ker(\phi) \leq G$ and $\text{im}(\phi) \leq H$.
5. If $K \leq G$ then $\phi(K) \leq H$.
6. If K is a cyclic subgroup of G then $\phi(K)$ is a cyclic subgroup of H .
7. If K is an abelian subgroup of G then $\phi(K)$ is an abelian subgroup of H .
8. If $L \leq H$ then $\phi^{-1}(L) \leq G$.
9. ϕ is injective if and only if $\ker(\phi) = \{e_G\}$.

Proof. These are all straightforward proofs and make excellent practice. We will do the proof of (8) and leave the others as exercises.

We will proceed using the subgroup test. Fix $a, b \in \phi^{-1}(L)$, which means that $\phi(a), \phi(b) \in L$ by definition. Since $L \leq H$ that means that $\phi(b)^{-1} \in L$ as well. Now

$$\begin{aligned} \phi(ab^{-1}) &= \phi(a)\phi(b^{-1}) && \phi \text{ a homomorphism} \\ &= \phi(a)\phi(b)^{-1} && \text{by property (2),} \end{aligned}$$

and since both of these are elements of L and L is a group, it follows that $\phi(ab^{-1}) \in L$. Thus $ab^{-1} \in \phi^{-1}(L)$, as required. □

Exercise 6 Prove the remaining properties from Proposition 5.

Isomorphism

Now that we have a better understanding of homomorphisms, it's time to realize our goal from the beginning of the readings: It's time to determine when two groups are secretly the same.

Definition 7

If G and H are two groups, we say that a group homomorphism $\phi : G \rightarrow H$ is an *isomorphism* if ϕ is also a bijection. If an isomorphism between G and H exists, then we say that G and H are *isomorphic* and write $G \cong H$.

Since homomorphisms already preserve the structure of the domain and translate it into the codomain, the additional bijectivity constraint is all that we need to say that the structure of G maps perfectly onto the structure of H . There is a perfect, one-to-one correspondence between the elements of G and H , and ϕ provides the translation between the two.

Exercise 8 Show that being isomorphic is an equivalence relation on the set of groups.

Example 9

Consider the groups $\mathbb{Z}_2$, U , and $\langle [2]_4 \rangle \leq \mathbb{Z}_4$ discussed at the beginning of the reading. Show that these groups are isomorphic.

Solution. Per Exercise 8, we know that being isomorphic is a transitive property, and hence it suffices to show that $\mathbb{Z}_2 \cong U$ and $\mathbb{Z}_2 \cong \langle [2]_4 \rangle$, from which $U \cong \langle [2]_4 \rangle$ will follow by transitivity. Each of these groups only has two elements, and since the identity elements must map to one another, the isomorphisms are evident. For the maps $\phi : \mathbb{Z}_2 \rightarrow U$ and $\psi : \mathbb{Z}_2 \rightarrow \langle [2]_4 \rangle$ define

$$\phi(x) = \begin{cases} [1]_6 & x = [0]_2 \\ [5]_6 & x = [1]_2 \end{cases} \quad \text{and} \quad \psi(x) = \begin{cases} [0]_4 & x = [0]_2 \\ [2]_4 & x = [1]_2 \end{cases}.$$

To see that ϕ is a homomorphism, we need only show that $\phi([1]_2 + [1]_2) = \phi([1]_2)\phi([1]_2)$ (why?). Indeed

$$\phi([1]_2 + [1]_2) = \phi([0]_2) = [1]_6 \quad \text{and} \quad \phi([1]_2)\phi([1]_2) = [5]_6 \cdot [5]_6 = [1]_6.$$

It is evidently bijective, since both the domain and codomain each only have two elements, and thus is an isomorphism. Precisely the same argument holds to show that ψ is a homomorphism and that it is bijective, so we leave it as an exercise. ■

Exercise 10 Complete Example 9 by showing that ψ is an isomorphism.

Exercise 11 Let D_3 be the dihedral group and r be the 120° rotation. Show that $\langle r \rangle \cong \mathbb{Z}_3$.

Similar to the properties of homomorphisms, we have the properties of isomorphisms below:

Proposition 12

Suppose that $\phi : G \rightarrow H$ is an isomorphism, so that G and H are isomorphic through ϕ .

1. $\phi^{-1} : H \rightarrow G$ exists and is an isomorphism.
2. G is abelian if and only if H is abelian.
3. G is cyclic if and only if H is cyclic.
4. If $K \leq G$ then $\phi(K) \cong K$.
5. For all $g \in G$, $|g| = |\phi(g)|$.
6. $|G| = |H|$.

Proof. Once again, these are straightforward and excellent practice. We will prove (2) and leave the rest as an exercise. Suppose that G and H are isomorphic, with $\phi : G \rightarrow H$ a corresponding isomorphism.

($\implies$) Suppose that G is abelian, and fix $h_1, h_2 \in H$. Since ϕ is bijective, there are $g_1, g_2 \in G$ such that $\phi(g_1) = h_1$ and $\phi(g_2) = h_2$. Now

$$\begin{aligned} h_1 h_2 &= \phi(g_1) \phi(g_2) = \phi(g_1 g_2) && \phi \text{ a homomorphism} \\ &= \phi(g_2 g_1) && G \text{ abelian} \\ &= \phi(g_2) \phi(g_1) = h_2 h_1 && \phi \text{ a homomorphism.} \end{aligned}$$

Thus H is abelian.

($\impliedby$) This follows from Property (1) and the proof of ($\implies$). $\square$

We end the course with the following theorem, which identifies for us that cyclic groups are unique up to their cardinality.

Theorem 13

Suppose G is a cyclic group.

1. If $|G| = \infty$ then $G \cong \mathbb{Z}$.
2. If $|G|$ is finite and $|G| = n$, then $G \cong \mathbb{Z}_n$.

Proof. We will prove (1) and leave (2) as an exercise. Fix a generator g of G , so that $G = \langle g \rangle$. Define $\phi : \mathbb{Z} \rightarrow G$ by $\phi(n) = g^n$, which we'll show is an isomorphism. We first show it is a homomorphism. Let $a, b \in \mathbb{Z}$, so that

$$\phi(a + b) = g^{a+b} = g^a g^b = \phi(a) \phi(b).$$

Next, we'll show that ϕ is injective by showing that its kernel is trivial. Note that if $n \in \ker(\phi)$ then $\phi(n) = g^n = e$. We claim that $n = 0$. Indeed, suppose that $n \neq 0$ and without loss of generality suppose that $n > 0$. By Exercise 25 in the Week 11 Readings, that means that $|G| = |\langle g \rangle| = n$, which contradicts the fact that $|G| = \infty$. Thus $n = 0$, and $\ker(\phi) \subseteq \{0\}$. The other inclusion is trivially true, and so $\ker(\phi) = \{0\}$, showing that ϕ is injective.

Finally, we need only show surjectivity. Let $h \in G$ be an arbitrary element. Since $G = \langle g \rangle$ is cyclic, we can write $h = g^n$ for some $n \in \mathbb{Z}$. Thus $\phi(n) = g^n = h$, showing that ϕ is surjective.

Thus ϕ is a bijective homomorphism and is thus an isomorphism, allowing us to conclude that $\mathbb{Z} \cong G$. $\square$

Exercise 14 Prove Theorem 13(2).

Exercises

1. Determine whether each of the following is a group homomorphism.
 - (a) Let $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ be a group under multiplication. Define $\phi : \mathbb{R}^* \rightarrow \mathbb{R}^*$ by $\phi(x) = x^2$.
 - (b) $\psi : \mathbb{Z}_4 \rightarrow \mathbb{Z}_4, \psi([n]_4) = [n+1]_4$.
 - (c) $\chi : \mathbb{Z} \rightarrow D_4, \chi(n) = r^n$.
 - (d) $\beta : S_3 \rightarrow \mathbb{Z}_3, \beta(\sigma) = |\{n \in \{1, 2, 3\} : \sigma(n) = n\}|$.
 - (e) $\gamma : S_3 \rightarrow S_3, \gamma(\sigma) = (12)\sigma$, where (12) is a cycle, in the same sense as the Week 11 Exercise Problems.
2. Determine the kernel for each of the following group homomorphisms. Be sure to prove your result.
 - (a) $\delta : \mathbb{Z} \rightarrow \mathbb{Z}_6, \delta(n) = [2n]_6$.
 - (b) $\rho : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_4, \rho([n]_{12}) = [n]_4$.
 - (c) $\tau : D_4 \rightarrow \mathbb{Z}_2, \tau(s) = \begin{cases} 0 & \text{if } s \text{ is a rotation} \\ 1 & \text{otherwise} \end{cases}$.
3. Suppose that $\phi : G \rightarrow K$ is a homomorphism with $\ker(\phi) = \{e_G\}$. Show that $G \cong \phi(G)$.
4. Let G be a group. Prove that $\phi : G \rightarrow G, g \mapsto g^{-1}$ is a homomorphism if and only if G is abelian.
5. Show that groups $\mathbb{R}$ and $\mathbb{Q}$ under addition are not isomorphic.
6. An isomorphism from a group to itself $\phi : G \xrightarrow{\cong} G$ is called an *automorphism*. For a fixed group G , let $\text{Aut}(G)$ denote the set of all automorphisms on G . Show that $\text{Aut}(G)$ is a group under function composition.
7. Let G be a group and $g \in G$ be a fixed element. The *conjugation map* is the map $c_g : G \rightarrow G$ given by $c_g(h) = ghg^{-1}$.
 - (a) Show that c_g is an automorphism of G for all $g \in G$.
 - (b) Let $\text{Inn}(G) = \{c_g : g \in G\}$. Show that $\text{Inn}(G)$ is a group under function composition.
 - (c) Conclude that $\text{Inn}(G) \leq \text{Aut}(G)$.
8. Consider the group $\mathbb{Q}$ under addition. Show that for any non-zero $q \in \mathbb{Q}$, the map $\phi_q : \mathbb{Q} \rightarrow \mathbb{Q}, \phi_q(r) = qr$ is an automorphism of $\mathbb{Q}$.